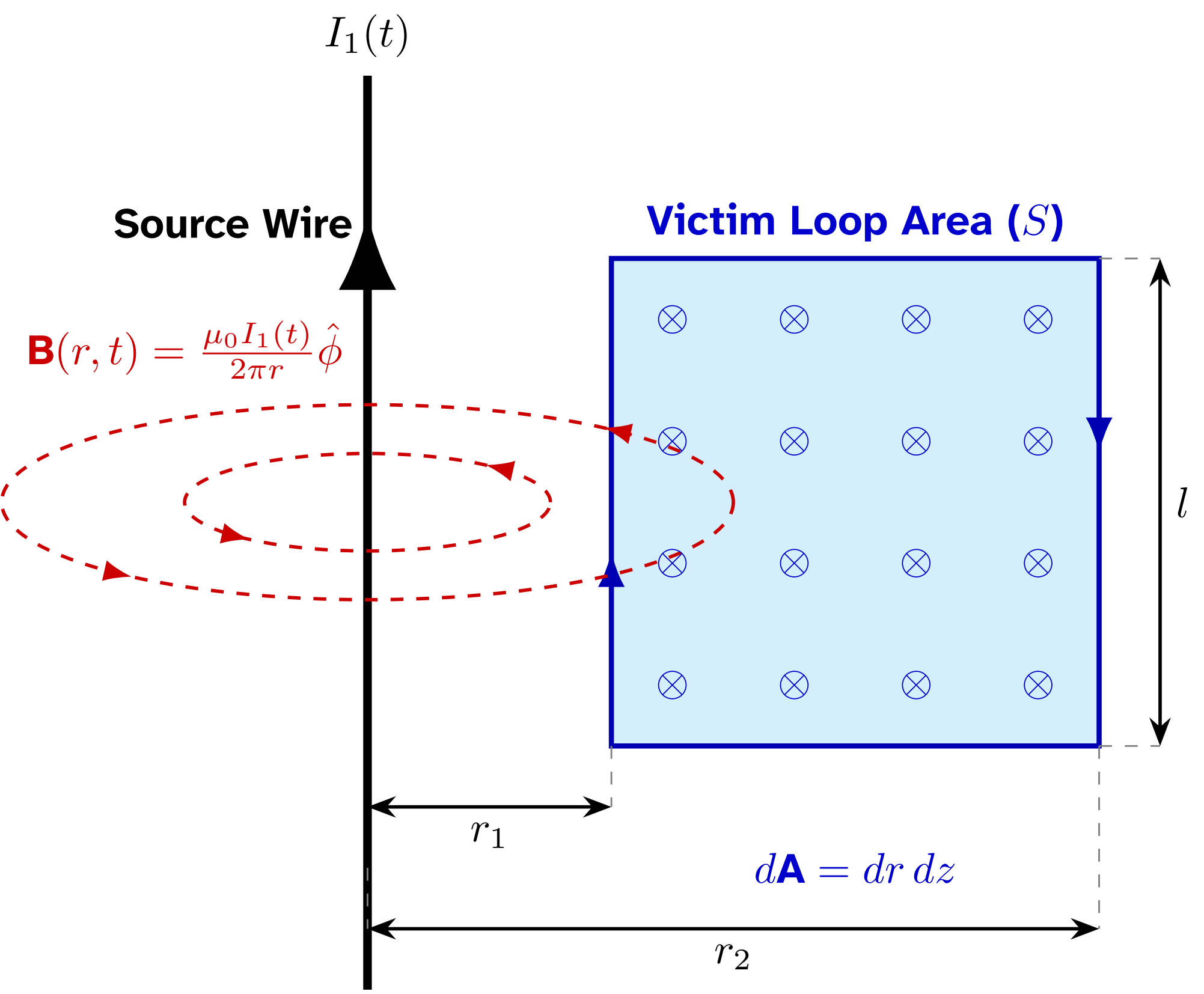

$I_1(t)$
Source Wire
Victim Loop Area (S)
$\mathbf{B}(r,t) = \frac{\mu_0 I_1(t)}{2\pi r}\hat{\phi}$
l
r_1
$d\mathbf{A} = dr\,dz$
r_2